Introduction to Software Engineering

Software Requirements Specification

The student team is required to complete the **Software Requirements Specification** documentation for the assigned course project, following the attached template.



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Software Requirements Specification

UniLib - University Library Management System

Team T05 | PA2 | Draft consolidated on 10 November 2026 | Human review and baseline approval pending

Objectives

This document focus on the following topics:

- ✔ Completing the Software Requirements Specification (SRS) document with the following sections:
  - Problem Statement
  - Stakeholders
  - Functional, non-functional and domain requirements
  - Use case model and use case specifications
  - Prototype / mockups
  - Traceability and requirements review
- ✔ Including an AI usage log for this phase
- ✔ Understanding the Software Requirements Specification (SRS) document.

1 Member Contribution Assessment

Provisional PA2 allocation: 25% per member (planned total: 100%). These are TARGET values from the project contribution framework, not independently verified actual effort. Final percentages and peer-review evidence require Team T05 approval at the formal SRS baseline review.

ID	Name	Role(s)	Contribution (%)	Evidence
23127520	Nguyen Phan Thi Vinh	Team Lead / PM; Tester	25% (target)	docs/pa2/00-contribution.md; GitHub issues #31, #32
24127469	Nguyen Thanh Thao Nguyen	Business Analyst	25% (target)	docs/pa2/02-glossary.md; 03-problem-statement.md; 05-functional-requirements.md; issues #20-27
24127324	Vu Thi Anh	Designer / Software Architect	25% (target)	pa/PA2/screen-inventory.md; Figma Reader/Admin/Librarian pages; design review pending
24127228	Tran Ho Anh Quan	Developer / Technical reviewer	25% (target)	docs/architecture/pa2-quan-technical-review.md; docs/ai-usage/pa2-q1-q6-quan.md

2 Introduction

2.1 Purpose and Scope

UniLib is a browser-based University Library Management System for a centralized small or medium-sized academic library. This SRS specifies what the proposed system shall do and how its behavior will be accepted; it does not claim that these functions have already been implemented. Intended readers are instructors and teaching assistants, student project team T05, representative Readers, Librarians, Administrators, and future PA3/PA4/PA5 designers and testers.

IN SCOPE: F01 Authentication and role-based authorization; F02 Book Title catalog management; F03 physical Book Copy inventory; F04 catalog search and browse; F05 checkout, check-in, overdue identification and loan renewal; F06 title-level reservations and FIFO hold processing. These six features are Must-have. F07 Reader dashboard and Librarian overdue console are Should-have, subject to schedule.

OUT OF SCOPE: e-book reading/DRM, financial payment or fine collection, physical RFID/barcode-scanner integrations, inter-library loans, AI recommendations, production email/SMS delivery, and native mobile apps. Manual Copy ID entry and in-app alerts remain in scope.

Source baseline: PA0/PA1 project documents, the frozen Master ID Registry (pa/PA2/00-id-registry.md), docs/pa2/*.md, the 1 Nov 2026 technical review, and the Figma design file. Changes after the registry freeze require recorded Project Manager approval.

2.2 Glossary

Term	Definition
Book Title	Bibliographic record identified by ISBN, including title, authors, category, publisher and publication year.
Book Copy	One physical item linked to a Book Title; identified by a unique Barcode / Copy ID.
ISBN	International Standard Book Number, unique for a Book Title within UniLib.
Barcode / Copy ID	Unique identifier of a physical copy; typed manually in the web UI; hardware integration excluded.
Reader	Authorized patron who searches the catalog, views personal loans, and places eligible reservations.
Librarian	Staff user managing Book Titles, copies, loans, returns, and reservation fulfillment.
Administrator	Privileged user managing accounts, role assignments, and account activation status.
RBAC	Role-Based Access Control: API operations authorized based on Reader/Librarian/Administrator roles.
Loan	Checkout transaction linking a Reader and physical Book Copy to a checkout date, due date and return.
Due Date	Return deadline set to 14 calendar days after checkout under BR-02.

Term	Definition
Loan Renewal	Once-per-loan 14-day extension, when not overdue and with no queued title reservation.
Overdue Loan	Unreturned active Loan for which the Due Date is in the past.
Reservation	Title-level request by an eligible Reader when no copies are currently Available.
FIFO Queue	First-In, First-Out ordering of Reservations by creation timestamp.
Ready for Pickup / Hold	Allocated Reserved copy awaiting pickup by the designated Reader.
Pickup Window	72-hour period following hold allocation before uncollected reservation expires.
Circulation Desk	Librarian-operated workflow for checkout, check-in and reservation fulfillment.
Available / Borrowed / Reserved	Core copy states: available for checkout; checked out; allocated to an eligible hold.
Damaged / Lost	Non-circulating physical copy condition/status in the approved model.
SRS / FR / NFR / BR / UC / SCR / AC	Specification / Functional Requirement / Non-Functional Requirement / Business Rule / Use Case / Screen / Acceptance Criterion.

3 Problem Statement

BUSINESS CONTEXT. University library staff need to locate, maintain and circulate physical academic books. In small academic collections, paper logbooks, spreadsheets and disconnected tools can leave Book Title metadata, physical-copy availability and circulation records inconsistent.

CURRENT PROBLEMS. (1) Readers cannot reliably inspect availability without contacting staff. (2) Staff must manually reconcile Book Copies against active Loans and may apply due dates, borrowing limits and overdue restrictions inconsistently. (3) Reservation requests may be handled without transparent FIFO order. (4) User privileges and account status may lack centralized control.

PROJECT GOALS. Deliver searchable bibliographic records and accurate copy counts; maintain one record per physical copy; process checked-out and returned copies with automatic 14-day dates and approved limits; enforce reservation eligibility, FIFO order and 72-hour holds; allow Admin account provisioning; and, if F07 is delivered, show personal/operational overdue information.

OPERATING ENVIRONMENT. Responsive web client using React/Vite, REST/JSON backend using Node.js 22 and Express, and PostgreSQL 16 database; GitHub for repository and project artifacts. Vercel/Render are proposed deployment platforms, not current production environments. Target desktop browsers: Chrome, Firefox and Edge 120+, Safari 17+, at 1024-1920 px viewport widths.

CONSTRAINTS. Must respect the frozen identifiers and BR-01 to BR-10; protect credentials and personal borrowing records; execute circulation changes atomically; keep scope within F01-F06 Must and F07 Should. The repository contains planning documents and an empty src/ placeholder, so runtime qualities and implemented behavior remain unverified.

OPEN ITEMS. Confirm canonical Reservation states, whether Overdue is derived or stored, whether Maintenance is a note or an official state, final screen links, and final baseline approval.

4 Stakeholders

The following five groups are identified in the project stakeholder analysis. Reader, Librarian and Administrator are primary software actors; course evaluators and Team T05 are governance / delivery stakeholders rather than application roles.

#	Stakeholder	Description	Interests / concerns
1	Project sponsor / evaluators	Course instructor and teaching assistants, HCMUS FIT.	Academic process, SRS quality, testable traceability, scope control and AI disclosure.
2	Reader / patron	Students, faculty and academic users of the library.	Fast search, accurate availability, fair reservations, privacy, loan due-date visibility.
3	Librarian	Circulation and cataloging staff.	Book/copy maintenance, correct checkout/return, automatic rules, hold handling.
4	Administrator	Library/IT account-management personnel.	User provisioning, role separation, account deactivation, credential protection.
5	Development & QA team	Team T05: PM/Tester, BA, Designer, Developer.	Clear baseline, feasible implementation, evidence-backed review, traceability.

5 Requirements

5.1 Functional Requirements

Requirements follow the frozen IDs in pa/PA2/00-id-registry.md. F01-F06 are Must-have; F07 is Should-have. All 32 statements are specifications, not evidence of implemented behavior. Source paths are retained for independent checking.

ID	Requirement	Feature	Priority	Source
FR-F01-01	The system shall authenticate registered users using username/email and password credentials.	F01	Must	PA0/PA1
FR-F01-02	The system shall issue a JWT on successful login and enforce Reader, Librarian and Administrator access permissions on protected API endpoints.	F01	Must	PA1; ID Registry
FR-F01-03	The system shall allow an authenticated user to log out by clearing the client-side session token.	F01	Must	ID Registry
FR-F01-04	The system shall allow Administrators to create accounts, assign roles, and activate/deactivate accounts.	F01	Must	PA0/PA1
FR-F02-01	The system shall allow Librarians to create Book Title records with unique ISBN, Title, Author(s), Category, Publisher and Publication Year.	F02	Must	PA0/PA1
FR-F02-02	The system shall allow Librarians to update Book Title metadata.	F02	Must	PA0/PA1
FR-F02-03	The system shall restrict Book Title removal/deactivation while copies, active loans or pending reservations are associated.	F02	Must	BR-09
FR-F02-04	The system shall display Total, Available, Borrowed and Reserved Book Copy counts for each title.	F02	Must	ID Registry
FR-F03-01	The system shall register copies under a Book Title with unique Barcode/Copy ID and Shelf Location.	F03	Must	PA0/PA1
FR-F03-02	The system shall allow Librarians to update copy Shelf Location and physical condition notes.	F03	Must	ID Registry
FR-F03-03	The system shall maintain approved copy states: Available, Borrowed, Reserved, Damaged and Lost.	F03	Must	ID Registry
FR-F03-04	The system shall permit decommissioning only for an Available or Damaged/Lost copy, never Borrowed/Reserved.	F03	Must	BR-09
FR-F04-01	The system shall search Book Titles by Title, Author, Category and ISBN keywords.	F04	Must	PA0/PA1
FR-F04-02	The system shall filter catalog results by Category and Availability and sort by Title or Publication Year.	F04	Must	ID Registry
FR-F04-03	The system shall display current Available copy counts and Shelf Locations on catalog results and details.	F04	Must	PA0/PA1

ID	Requirement	Feature	Priority	Source
FR-F04-04	The system shall provide server-side paginated catalog browsing with a default of 10 titles per page.	F04	Must	ID Registry
FR-F05-01	The system shall allow a Librarian to create a Loan for an active Reader and an Available Copy ID.	F05	Must	PA0/PA1
FR-F05-02	The system shall reject checkout unless Reader is active, has fewer than 3 active Loans and no Overdue Loan.	F05	Must	BR-01; BR-04
FR-F05-03	The system shall calculate Due Date as checkout plus 14 calendar days and transition the copy to Borrowed.	F05	Must	BR-02
FR-F05-04	The system shall record return timestamp, close the active Loan and process copy check-in.	F05	Must	PA0/PA1
FR-F05-05	The system shall allocate a returned copy to the first eligible FIFO Reservation as Reserved or otherwise make it Available.	F05	Must	BR-07; BR-08
FR-F05-06	The system shall permit one 14-day renewal when Loan is not Overdue and there is no pending Reservation for its title.	F05	Must	BR-03
FR-F05-07	The system shall identify and flag active Loans past their Due Dates as Overdue.	F05	Must	PA0/PA1
FR-F06-01	The system shall allow a Reader to reserve a title only when zero associated copies are Available.	F06	Must	BR-05
FR-F06-02	The system shall reject reservation if Reader has 2 active Reservations or an active Loan for that title.	F06	Must	BR-05; BR-06
FR-F06-03	The system shall place same-title Reservations in FIFO order by creation timestamp.	F06	Must	BR-07
FR-F06-04	The system shall assign a 72-hour Pickup Window on allocation of a copy to a Reservation.	F06	Must	BR-08
FR-F06-05	The system shall support cancellation of pending Reservations and expiry of uncollected holds, advancing FIFO or releasing copy.	F06	Must	BR-08
FR-F06-06	The system shall fulfill a held Reservation when its designated Reader checks out the allocated copy.	F06	Must	PA0/PA1
FR-F07-01	The system shall present Reader active Loans, Due Dates, renewal actions, Reservations and queue positions in a dashboard.	F07	Should	PA1
FR-F07-02	The system shall display Overdue warnings and borrowing restriction notices to Readers.	F07	Should	PA1
FR-F07-03	The system shall provide Librarians an Overdue desk with late days, Reader contacts and Book Copy details.	F07	Should	PA1

5.2 Non-Functional Requirements

The following measurable Non-Functional Requirements define performance benchmarks, security standards, usability criteria, and system quality attributes:

ID	Category	Requirement (measurable)	How it will be verified

NFR-01	Performance	Under 100 concurrent simulated users, catalog search queries shall return response payloads in <= 1.0 s at P95 across 10,000 book records.	Automated Load Test (k6 / Artillery)
NFR-02	Performance	Initial application load time shall achieve Largest Contentful Paint (LCP) <= 2.5 s at P90 on standard broadband/4G profiles.	Lighthouse Performance Audit
NFR-03	Security	User passwords shall be hashed using bcrypt with salt rounds >= 10; plain-text credentials shall never be logged or transmitted in response payloads.	Code Review & Unit Test Inspection
NFR-04	Security	All protected REST endpoints shall validate JWT claims; unauthenticated requests return HTTP 401; unauthorized roles return HTTP 403.	Automated API Integration Tests (Jest + Supertest)
NFR-05	Data Integrity	Circulation state changes (checkout, return, hold allocation) shall execute within atomic DB transactions with row-level locks, guaranteeing 0 race conditions.	Concurrency Integration Test Suite
NFR-06	Usability	The user interface shall achieve Google Lighthouse Accessibility score >= 85, color contrast ratio >= 4.5:1, and support keyboard navigation.	Lighthouse Accessibility Audit & Manual Walkthrough
NFR-07	Compatibility	The web application shall render correctly on modern evergreen browsers (Chrome 120+, Firefox 120+, Edge 120+, Safari 17+) on viewports 1024px to 1920px.	Cross-browser Verification & Viewport Emulation
NFR-08	Reliability	The system shall display user-friendly error banners upon network/API failures, and support database schema restoration via seed scripts in < 5 minutes.	Fault Injection Test & Recovery Run

5.3 Domain Requirements and Business Rules

ID	Rule	Source (policy, law, standard)
BR-01	Maximum Active Loan Limit: A Reader is strictly limited to a maximum of 3 concurrently active (unreturned) book loans across the entire library system.	Library Circulation Policy §2.1 (Kickoff 23/10)
BR-02	Standard Loan Period: The default borrowing duration is 14 calendar days from checkout timestamp. Due date is calculated automatically at checkout.	Academic Calendar & Loan Rules §3.2 (Kickoff 23/10)
BR-03	Loan Renewal Policy: A loan may be renewed exactly once for an additional 14 calendar days, provided the loan is not overdue and has 0 title reservations.	Circulation Renewal Policy §3.4
BR-04	Overdue Borrowing Suspension: Any Reader possessing at least one overdue loan is immediately blocked from new checkouts and new reservations until overdue items are returned.	Circulation Sanction Rule §4.1
BR-05	Reservation Eligibility: A Reader may place a reservation for a book title if and only if zero physical copies are currently Available, and reader has no active loan of the title.	Reservation Policy §5.1
BR-06	Maximum Active Reservations: A Reader may hold a maximum of 2 active reservations concurrently across the system.	Reservation Policy §5.2

BR-07	Reservation Priority Order: Book reservations are queued and processed in strict First-In, First-Out (FIFO) chronological sequence based on reservation timestamp.	Reservation Queue Protocol §5.3
BR-08	Reservation Hold Pickup Window: When a returned copy is routed to fulfill a reservation, it is held with status Reserved for exactly 3 calendar days (72 hours) before auto-expiry.	Hold Shelf Policy §5.4 (Kickoff 23/10)
BR-09	Deletion Referential Integrity: A book title cannot be deleted if physical copies exist. A physical copy cannot be removed if current status is Borrowed or Reserved.	Database Integrity Standard & Catalog Policy §6.1
BR-10	Separation of Roles: Readers have read-only access to catalog/personal data; Librarians manage circulation and catalog; Administrators manage user credentials.	Security Access Policy §1.3

5.4 Acceptance Criteria

Each AC is a testable Given/When/Then scenario. The 30 AC IDs below follow docs/pa2/08-acceptance-criteria.md. Verification outcomes are planned for PA4/PA5, not claimed as passed.

AC ID	Feature	Given / When / Then	Related FR
AC-F01-01	F01	Given an active user with valid credentials; When the user logs in; Then issue a JWT and show the authorized role-specific landing page.	FR-F01-01/02
AC-F01-02	F01	Given invalid credentials; When login is submitted; Then reject the session with a generic error and no token.	FR-F01-01
AC-F01-03	F01	Given a Reader token; When a protected staff endpoint is requested; Then return HTTP 403 and make no change.	FR-F01-02
AC-F01-04	F01	Given a deactivated account; When credentials are submitted; Then reject login and do not issue a session.	FR-F01-04
AC-F02-01	F02	Given a Librarian and a unique ISBN; When a complete title is saved; Then create it and display copy count 0.	FR-F02-01/04
AC-F02-02	F02	Given an existing ISBN; When a duplicate is created; Then reject it and preserve the existing title.	FR-F02-01
AC-F02-03	F02	Given a title with a copy; When deletion is attempted; Then reject deletion and preserve the title.	FR-F02-03
AC-F02-04	F02	Given a Reader; When a catalog write is attempted; Then deny access and make no change.	FR-F02-01/02
AC-F03-01	F03	Given a title and unused barcode; When a Librarian registers a copy; Then mark it Available and increment copy count.	FR-F03-01/03

AC ID	Feature	Given / When / Then	Related FR
AC-F03-02	F03	Given a Borrowed/Reserved copy; When decommission is requested; Then reject it and retain status.	FR-F03-04
AC-F03-03	F03	Given a Reader; When copy metadata is changed; Then return HTTP 403 and keep inventory unchanged.	FR-F03-01
AC-F03-04	F03	Given an existing Copy ID; When the same ID is registered; Then reject duplicate registration.	FR-F03-01
AC-F04-01	F04	Given 10,000 seeded titles; When search is executed under 100 concurrent users; Then P95 response <= 1 s and copy counters appear.	FR-F04-01/03
AC-F04-02	F04	Given mixed results; When Category and Available Now filters are selected; Then show only matching available titles.	FR-F04-02
AC-F04-03	F04	Given a query with no matches; When search completes; Then show an empty state without an error.	FR-F04-01
AC-F04-04	F04	Given >10 matching titles; When page 1 and 2 are requested; Then return 10 per page with no duplicate record.	FR-F04-04
AC-F05-01	F05	Given an eligible Reader and Available Copy; When Librarian checks out; Then create a Loan, +14-day Due Date and Borrowed state.	FR-F05-01/03
AC-F05-02	F05	Given 3 active Loans; When a fourth is requested; Then reject without changing copy or loan count.	FR-F05-02
AC-F05-03	F05	Given an Overdue Loan; When checkout is requested; Then reject and leave the copy Available.	FR-F05-02
AC-F05-04	F05	Given active Loan with no holds; When copy is returned; Then close Loan and mark copy Available.	FR-F05-04/05
AC-F05-05	F05	Given a pending title queue; When copy returns; Then reserve it for oldest eligible request for 72 hours.	FR-F05-04/05
AC-F05-06	F05	Given a non-overdue, never-renewed Loan with no holds; When renewed; Then extend 14 days once and reject second renewal.	FR-F05-06
AC-F05-07	F05	Given Reader-only permissions; When checkout/check-in API is called; Then return HTTP 403 and change nothing.	FR-F05-01/04
AC-F06-01	F06	Given no Available copies and eligible Reader; When reserve is submitted; Then create Pending hold at FIFO tail.	FR-F06-01/03

AC ID	Feature	Given / When / Then	Related FR
AC-F06-02	F06	Given an Available copy; When reserve is requested; Then reject the request.	FR-F06-01
AC-F06-03	F06	Given 2 active Reservations; When a third is requested; Then reject with no additional hold.	FR-F06-02
AC-F06-04	F06	Given a Ready hold uncollected for 72 h; When expiry processing runs; Then expire it and advance FIFO or release copy.	FR-F06-04/05
AC-F06-05	F06	Given a Reader currently borrowing title X; When X is reserved; Then reject the duplicate-title request.	FR-F06-02
AC-F07-01	F07	Given Reader loans, holds and overdue flags; When dashboard opens; Then display corresponding due dates, queues and warnings.	FR-F07-01/02
AC-F07-02	F07	Given overdue Loans; When Librarian opens monitoring; Then list them with overdue-day counts and Reader details.	FR-F07-03

6 Use Case Model

6.1 Use Case Diagram

The use-case model below represents role-to-goal associations. Authentication is an independent use case; no include/extend/generalization dependency is asserted without an approved business justification. See the registry (UC-01 to UC-12) for authoritative IDs.

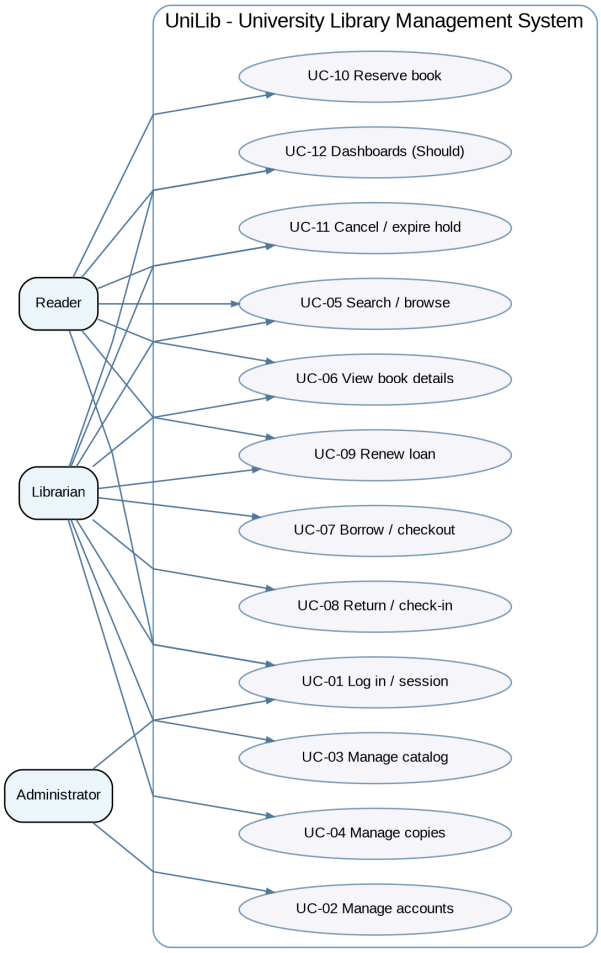


Figure 1. UniLib use-case model (derived from the frozen UC registry).

6.2 Actors

Actor	Description
Reader	Registered patron: logs in, searches the catalog, views availability, reserves/renews eligible items and tracks own circulation.
Librarian	Authorized staff member: maintains titles/copies, processes checkout/returns, routes holds and monitors overdue records.
Administrator	Authorized system manager: creates/updates user accounts, assigns roles and manages active/inactive status.
System scheduler (supporting)	Automated time-triggered hold-expiry operation; not an independently authenticated human actor.

6.3 Use Case List

ID	Use case	Primary actor	Priority	Related FR
UC-01	Log In & Manage Session	Reader, Librarian, Admin	Must*	FR-F01-01, FR-F01-02, FR-F01-03
UC-02	Manage User Accounts	Administrator	Must	FR-F01-04
UC-03	Manage Book Catalog (CRUD Titles)	Librarian	Must*	FR-F02-01, FR-F02-02, FR-F02-03, FR-F02-04
UC-04	Manage Book Copies & Inventory	Librarian	Must	FR-F03-01, FR-F03-02, FR-F03-03, FR-F03-04
UC-05	Search & Browse Book Catalog	Reader, Librarian	Must*	FR-F04-01, FR-F04-02, FR-F04-03, FR-F04-04
UC-06	View Book Details & Copy Availability	Reader, Librarian	Must	FR-F04-03, FR-F02-04
UC-07	Borrow Book (Circulation Checkout)	Librarian	Must*	FR-F05-01, FR-F05-02, FR-F05-03
UC-08	Return Book (Circulation Check-in)	Librarian	Must*	FR-F05-04, FR-F05-05
UC-09	Renew Book Loan	Reader, Librarian	Must	FR-F05-06, BR-03
UC-10	Reserve Book Title (Place Hold)	Reader	Must*	FR-F06-01, FR-F06-02, FR-F06-03
UC-11	Cancel Reservation & Process Expiry	Reader, System / Librarian	Must	FR-F06-04, FR-F06-05, BR-08
UC-12	View Reader Dashboard & Overdue Desk	Reader, Librarian	Should	FR-F07-01, FR-F07-02, FR-F07-03

7 Use Case Specifications

All 11 Must-have use cases plus the Should-have UC-12 are specified below. Steps are numbered; each scenario references the frozen registry and remains subject to BA/team approval.

7.1 Use Case UC-01

Use case ID	UC-01
Use case name	Log In & Manage Session
Brief description	Authenticate an authorized user via username/email and password credentials and establish a secure role-based session token.
Primary actor	Reader, Librarian, Administrator
Secondary actors	Authentication Service / Database Service
Priority	Must*
Related requirements	FR-F01-01, FR-F01-02, FR-F01-03
Trigger	User navigates to the Login page (SCR-01) or attempts to access a protected URL.
Pre-conditions	User account has been registered by the Administrator and is currently in Active status.
Post-conditions	A cryptographically signed JWT with role claim is issued; user is redirected to their role landing dashboard; or session is terminated upon logout.
Main flow	1. User opens the login interface (SCR-01). 2. User enters username/email and password, then submits credentials. 3. System validates input formats and hashes password with bcrypt (salt >= 10). 4. System verifies credentials against the user database and checks active status. 5. System generates signed JWT containing user ID and role claim. 6. System redirects user to their role landing page (Reader: SCR-11, Librarian: SCR-07, Admin: SCR-02).
Alternative flows	A1 (at step 2): User toggles password visibility. A2 (at step 6): User clicks Logout; system clears client session token and redirects to SCR-01.
Exception flows	E1 (at step 4): Invalid username or password; system displays "Invalid credentials" error banner without revealing which field failed (NFR-03). E2 (at step 4): Account is deactivated; system displays "Account suspended, contact Administrator" (FR-F01-04).
Business rules	BR-10 (Separation of Roles)
Non-functional constraints	NFR-03 (Password Hashing bcrypt >= 10), NFR-04 (JWT Authorization, 401/403 handling)

7.2 Use Case UC-02

Use case ID	UC-02
Use case name	Manage User Accounts

Brief description	Provision and manage role-based library user accounts.
Primary actor	Administrator
Secondary actors	System scheduler / database service (where applicable)
Priority	Must
Related requirements	FR-F01-04
Trigger	Administrator chooses User Management.
Pre-conditions	Administrator authenticated; privileged permission.
Post-conditions	Account created/updated or deactivated; audit evidence preserved.
Main flow	1. Open user management list. 2. Filter/search for a user or select Add User. 3. Enter account identity and assign Reader/Librarian/Administrator role. 4. System validates unique account identifiers and role choice. 5. Save account and show resulting state. 6. Administrator may activate/deactivate an existing account.
Alternative flows	A1 (step 3): Edit a current account instead of creating one.
Exception flows	E1 (step 4): Duplicate/invalid identity is rejected; E2: non-Admin forbidden.
Business rules	BR-10
Non-functional constraints	NFR-03, NFR-04

7.3 Use Case UC-03

Use case ID	UC-03
Use case name	Manage Book Catalog (CRUD Titles)
Brief description	Create, update, and manage bibliographic book title records in the catalog while maintaining referential integrity.
Primary actor	Librarian
Secondary actors	Database Service
Priority	Must*
Related requirements	FR-F02-01, FR-F02-02, FR-F02-03, FR-F02-04
Trigger	Librarian accesses Catalog Management Console (SCR-03).
Pre-conditions	Authenticated user with Librarian role.
Post-conditions	Book title record is created or updated in the catalog database; copy statistics recalculated.
Main flow	1. Librarian opens Catalog Management (SCR-03). 2. Librarian clicks "Add New Book" and enters ISBN, Title, Author(s), Category, Publisher, and Publication Year. 3. System validates ISBN uniqueness and metadata field formats. 4. System persists new book title and renders updated catalog table with copy count badge = 0. 5. Librarian may edit metadata of existing titles or inspect copy statistics.
Alternative flows	A1 (at step 2): Librarian edits existing title metadata; system persists changes and updates display.

	A2 (at step 5): Librarian requests title deletion; system verifies 0 physical copies and 0 active holds, then removes record.
Exception flows	E1 (at step 3): ISBN already exists; system rejects submission with duplicate ISBN error (FR-F02-01). E2 (at step A2): Title has linked physical copies or active loans/holds; system blocks deletion per BR-09.
Business rules	BR-09 (Deletion Referential Integrity)
Non-functional constraints	NFR-04 (Librarian RBAC Authorization), NFR-08 (Reliable persistence)

7.4 Use Case UC-04

Use case ID	UC-04
Use case name	Manage Book Copies & Inventory
Brief description	Maintain individual physical inventory units under a bibliographic title.
Primary actor	Librarian
Secondary actors	System scheduler / database service (where applicable)
Priority	Must
Related requirements	FR-F03-01 to 04
Trigger	Librarian opens Book Copy Inventory.
Pre-conditions	Librarian authenticated; parent title exists.
Post-conditions	Unique copy record created/updated with approved state.
Main flow	1. Select Book Title and open its copy list. 2. Choose Add Copy or Edit Copy. 3. Enter unique Barcode/Copy ID, shelf location and condition notes. 4. System validates unique identifier and permitted copy-state transition. 5. Save inventory update and refresh title copy counters. 6. For decommission, check copy not Borrowed or Reserved.
Alternative flows	A1 (step 2): Update shelf location or condition of an existing copy.
Exception flows	E1 (step 4): Duplicate barcode rejected; E2 (step 6): active copy cannot be decommissioned.
Business rules	BR-09, BR-10
Non-functional constraints	NFR-04, NFR-05

7.5 Use Case UC-05

Use case ID	UC-05
Use case name	Search & Browse Book Catalog
Brief description	Search the library catalog by keywords and filter by category or availability status with real-time copy counts.
Primary actor	Reader, Librarian
Secondary actors	Search Indexer / Database Service

Priority	Must*
Related requirements	FR-F04-01, FR-F04-02, FR-F04-03, FR-F04-04
Trigger	User navigates to Catalog Search view (SCR-05) or enters search terms in header bar.
Pre-conditions	System catalog is initialized and available.
Post-conditions	Matching book titles are displayed with real-time available copy counts and shelf locations.
Main flow	1. User accesses Search & Browse interface (SCR-05). 2. User enters search keywords (matching Title, Author, Category, or ISBN). 3. User applies optional filters (Category, Availability: "Available Now") and sorting (Title A-Z, Year). 4. System queries catalog index and returns paginated results (default 10 items/page) within <= 1.0s (NFR-01). 5. System renders book cards showing cover, bibliographic metadata, shelf location, and available copy counter. 6. User clicks a book card to view detailed copy inventory (SCR-06).
Alternative flows	A1 (at step 2): User leaves search bar empty; system displays paginated list of all active catalog titles. A2 (at step 3): User switches pagination page; system fetches next batch via server-side pagination.
Exception flows	E1 (at step 4): No catalog titles match search query; system displays friendly "No books found" message with filter reset suggestion.
Business rules	BR-10 (Public / Reader Access)
Non-functional constraints	NFR-01 (Search Latency <= 1.0s at P95 under 100 VU across 10,000 records), NFR-02 (LCP <= 2.5s)

7.6 Use Case UC-06

Use case ID	UC-06
Use case name	View Book Details & Availability
Brief description	Inspect title metadata and status/shelf of every physical copy.
Primary actor	Reader / Librarian
Secondary actors	System scheduler / database service (where applicable)
Priority	Must
Related requirements	FR-F04-03; FR-F02-04
Trigger	User selects book from Catalog.
Pre-conditions	Book Title exists.
Post-conditions	Complete title detail with aggregate and per-copy availability shown.

Main flow	1. Open a selected catalog result. 2. System loads metadata: ISBN, title, author, category, publisher and year. 3. System computes Total/Available/Borrowed/Reserved copy counts. 4. System lists Copy ID, shelf location and current state. 5. System offers Reserve only when the Reader meets reservation eligibility.
Alternative flows	A1 (step 5): User browses details without making a reservation.
Exception flows	E1: Book Title deleted/unknown shows not-found state; no stale counts displayed.
Business rules	BR-05
Non-functional constraints	NFR-01, NFR-06

7.7 Use Case UC-07

Use case ID	UC-07
Use case name	Borrow Book (Circulation Checkout)
Brief description	Librarian issues a book loan to an eligible Reader, calculating due date and transitioning copy to Borrowed.
Primary actor	Librarian
Secondary actors	Database Service
Priority	Must*
Related requirements	FR-F05-01, FR-F05-02, FR-F05-03, FR-F06-06
Trigger	Reader presents library card/ID and physical book copy at the circulation desk (SCR-07).
Pre-conditions	Authenticated Librarian; Reader account is active; copy is physically present.
Post-conditions	Active loan record created with due date = today + 14 days; copy status updated to Borrowed.
Main flow	1. Librarian opens Circulation Desk - Borrow (SCR-07). 2. Librarian enters Reader ID; system retrieves patron profile and verifies eligibility: < 3 active loans (BR-01) and 0 overdue loans (BR-04). 3. Librarian enters/scans Copy Barcode ID; system verifies copy status is Available. 4. System locks copy row with SELECT ... FOR UPDATE (NFR-05) within a database transaction. 5. System computes due date = checkout date + 14 calendar days (BR-02). 6. System creates active Loan record and transitions Copy status to Borrowed. 7. System displays checkout confirmation banner with due date and updated patron loan count.
Alternative flows	A1 (at step 3): Copy is currently Reserved on hold shelf for this specific Reader; system fulfills reservation (FR-F06-06) and converts hold into active loan.
Exception flows	E1 (at step 2): Reader has 3 active loans; system blocks checkout with "Maximum loan limit (3 books) reached" per BR-01. E2 (at step 2): Reader has >= 1 overdue loan; system blocks checkout with "Borrowing suspended: overdue loan present" per BR-04.

	E3 (at step 3): Copy status is Borrowed or Damaged; system blocks checkout with state conflict error.
Business rules	BR-01 (Max 3 Loans), BR-02 (14-day Duration), BR-04 (Overdue Suspension), BR-10 (Librarian RBAC)
Non-functional constraints	NFR-05 (Transactional Atomic Checkout, 0 Race Conditions)

7.8 Use Case UC-08

Use case ID	UC-08
Use case name	Return Book (Circulation Check-in)
Brief description	Librarian processes a returned book copy, recording check-in timestamp, assessing condition, and routing to hold shelf or available.
Primary actor	Librarian
Secondary actors	Database Service / Reservation Scheduler
Priority	Must*
Related requirements	FR-F05-04, FR-F05-05
Trigger	Patron returns physical book copy to circulation desk (SCR-08).
Pre-conditions	Authenticated Librarian; copy was checked out on an active loan.
Post-conditions	Loan status marked Returned; copy status transitions to Reserved (if holds exist) or Available.
Main flow	1. Librarian opens Circulation Desk - Return (SCR-08). 2. Librarian enters/scans Copy Barcode ID. 3. System locates active Loan record and locks copy row in database transaction (NFR-05). 4. System records return timestamp and marks Loan status as Returned. 5. System evaluates whether pending reservations exist for the book title in FIFO queue (BR-07). 6. If zero pending reservations: system transitions copy status to Available and directs shelving. 7. System displays check-in confirmation toast with shelving destination.
Alternative flows	A1 (at step 5): Pending reservation exists; system transitions copy status to Reserved, allocates copy to top FIFO patron, initiates 72-hour pickup window (BR-08), and displays hold routing notice.
Exception flows	E1 (at step 2): Copy ID does not correspond to an active loan; system alerts Librarian "No active loan found for this copy". E2 (at step 4): Copy is returned in damaged condition; Librarian flags condition; copy transitions to Damaged/Lost (never routed to hold shelf per TA-FB-02) and pending reservation remains in queue.
Business rules	BR-07 (FIFO Queue), BR-08 (72h Hold Window), BR-09 (Integrity)
Non-functional constraints	NFR-05 (Atomic Check-in and Hold Routing)

7.9 Use Case UC-09

Use case ID	UC-09
Use case name	Renew Book Loan
Brief description	Extend an eligible Loan exactly once.
Primary actor	Reader / Librarian
Secondary actors	System scheduler / database service (where applicable)
Priority	Must
Related requirements	FR-F05-06
Trigger	Reader or Librarian requests Renew.
Pre-conditions	Active Loan; not overdue; no pending title reservation; not previously renewed.
Post-conditions	Due Date extended by 14 days and renewal recorded.
Main flow	1. Open active Loan details or checkout screen. 2. Choose Renew. 3. System locks Loan and checks overdue, renewal count and pending title holds. 4. System extends Due Date by 14 calendar days. 5. System records renewal and displays revised date.
Alternative flows	A1 (step 1): Librarian submits renewal on behalf of Reader.
Exception flows	E1 (step 3): Overdue, existing hold or already renewed -> reject with reason.
Business rules	BR-03, BR-04
Non-functional constraints	NFR-04, NFR-05

7.10 Use Case UC-10

Use case ID	UC-10
Use case name	Reserve Book Title (Place Hold)
Brief description	Reader places a title-level reservation when zero copies are available, entering the FIFO queue.
Primary actor	Reader
Secondary actors	Reservation Queue Service
Priority	Must*
Related requirements	FR-F06-01, FR-F06-02, FR-F06-03
Trigger	Reader clicks "Reserve Book" button on Book Detail view (SCR-06).
Pre-conditions	Authenticated Reader; zero copies of book title are Available (BR-05); Reader has < 2 active reservations (BR-06); Reader has 0 overdue loans (BR-04) and no active loan of this title.
Post-conditions	Reservation record created with timestamp and queue position; reader informed of hold policy.

Main flow	1. Reader views Book Detail (SCR-06) showing 0 Available copies. 2. Reader clicks "Reserve Book" action button. 3. System displays Reservation Confirmation Modal (SCR-09) showing estimated queue position and 72-hour pickup policy disclaimer. 4. Reader confirms reservation request. 5. System verifies reservation constraints in database transaction: < 2 active holds, no active loan for title, no overdue items. 6. System appends reservation to title FIFO queue with creation timestamp (BR-07). 7. System displays success alert and updates Reader Personal Dashboard (SCR-11).
Alternative flows	A1 (at step 4): Reader cancels modal; system closes modal without recording hold.
Exception flows	E1 (at step 1): At least 1 copy is Available; system disables Reserve button and prompts reader to borrow available copy (BR-05). E2 (at step 5): Reader already has 2 active reservations; system rejects request with "Maximum active reservations (2) reached" (BR-06). E3 (at step 5): Reader currently has overdue loan; system blocks reservation per BR-04.
Business rules	BR-04 (Overdue Suspension), BR-05 (Reservation Eligibility), BR-06 (Max 2 Holds), BR-07 (FIFO Queue)
Non-functional constraints	NFR-04 (Reader RBAC), NFR-05 (Atomic Hold Allocation)

7.11 Use Case UC-11

Use case ID	UC-11
Use case name	Cancel Reservation & Process Expiry
Brief description	Cancel a Reader Reservation or expire a missed pickup window.
Primary actor	Reader / System
Secondary actors	System scheduler / database service (where applicable)
Priority	Must
Related requirements	FR-F06-04, 05
Trigger	Reader clicks Cancel, or 72-hour deadline elapses.
Pre-conditions	Reservation is Pending/Ready for Pickup.
Post-conditions	Reservation Cancelled/Expired; next FIFO reader gets a hold or copy returns Available.
Main flow	1. User opens My Reservations, or system detects expired deadline. 2. System locates and locks the current Reservation and allocated copy if any. 3. For user cancellation, system verifies ownership and cancelable state.

	4. System updates status to Cancelled or Expired. 5. If a reserved copy was released, allocate it to next FIFO eligible Reader with new 72-hour deadline. 6. If queue is empty, mark copy Available and publish updated status.
Alternative flows	A1 (step 1): System expiry runs without a Reader action.
Exception flows	E1 (step 3): Unauthorized/stale cancel rejected; E2: concurrent allocation rolls back and retries.
Business rules	BR-07, BR-08
Non-functional constraints	NFR-04, NFR-05

7.12 Use Case UC-12

Use case ID	UC-12
Use case name	View Reader Dashboard & Overdue Desk
Brief description	Reader views personal borrowings and holds; Librarian monitors overdue loans across the library.
Primary actor	Reader, Librarian
Secondary actors	Database Service
Priority	Should
Related requirements	FR-F07-01, FR-F07-02, FR-F07-03
Trigger	User navigates to My Dashboard (SCR-11) or Overdue Monitoring Console (SCR-12).
Pre-conditions	Authenticated user with Reader or Librarian role.
Post-conditions	Role-appropriate circulation summaries, due dates, overdue badges, and action triggers rendered.
Main flow	1. Authenticated user navigates to their circulation monitoring view. 2. System queries active loans, calculated due dates, and pending reservations for the user or library. 3. For Reader (SCR-11): system renders active loans table, due-date progress bars, eligible Renew button, active holds list, and prominent Overdue warning banner if overdue items exist (FR-F07-02). 4. For Librarian (SCR-12): system renders delinquent loans table sorted by days overdue with reader contact details and copy IDs (FR-F07-03). 5. User reviews records or initiates eligible actions (e.g. renewal or patron contact).
Alternative flows	A1 (at step 4): Librarian filters overdue list by days overdue (> 7 days, > 14 days) or exports summary view.
Exception flows	E1 (at step 2): Database/network timeout occurs; system displays user-friendly error banner without crashing (NFR-08).
Business rules	BR-03 (Renewal Policy), BR-04 (Overdue Suspension), BR-10 (Separation of Roles)
Non-functional constraints	NFR-04 (RBAC Protection), NFR-06 (Accessibility, WCAG Contrast >= 4.5:1), NFR-08 (Graceful Error Handling)

8 Prototype / Mockups

Primary interactive Figma source: https://www.figma.com/design/VGG2HSwsJmU0uNHUAjmvdZ/hehe--Copy- . The file has Reader, Administrator and Librarian pages. Links below point to identified top-level frames, not exported production screens. Screens SCR-09 and SCR-10 have no clearly identified dedicated top-level frame in the inspected page inventory; only related Reader reservation interactions are linked. Accessibility targets are specified (contrast 4.5:1, labels, keyboard operation), but no Figma-based accessibility audit has been performed.

Screen	Related use case(s)	Mockup / link
SCR-01 Authentication & Login	UC-01	Open 20:6778 | Reader Login; also Admin 21:12113 and Librarian 21:11799
SCR-02 Admin User Management	UC-02	Open 40:14803 | Admin / User Management
SCR-03 Catalog Management	UC-03	Open 2:2162 | Librarian / Book Catalog Management
SCR-04 Copy Inventory	UC-04	Open 2:2774 | Librarian / Book Copy Inventory
SCR-05 Catalog Search	UC-05	Open 2:672 | Reader / Catalog
SCR-06 Book Details	UC-06, UC-10	Open 40:15875 | Reader / Book details (Book 1; others also present)
SCR-07 Borrow Book	UC-07	Open 2:4163 | Librarian / Borrow Book
SCR-08 Return Book	UC-08	Open 2:4579 | Librarian / Return Book
SCR-09 Reservation Confirmation	UC-10	Open 40:15875 | Related Book Details Reserve action; dedicated modal not confirmed
SCR-10 Hold Shelf / Queue	UC-10, UC-11	Dedicated Librarian hold-shelf frame not identified; design gap
SCR-11 Reader Dashboard	UC-09, UC-11, UC-12	Open 2:242 | Reader / My Dashboard; related reservation variants
SCR-12 Overdue Monitoring	UC-12	Open 2:4933 | Librarian / Overdue & Circulation Monitoring

9 Requirements Traceability Matrix

The forward traceability matrix establishes links from functional requirements to use cases, wireframe screens, PA3 architecture design elements, and PA4 test cases:

Req. ID	Use case	Screen	Design element (PA3)	Test cases (PA4)
FR-F01-01	UC-01	SCR-01	[TBD: PA3 AuthController / POST /auth/login]	[TBD: PA4 TC-F01-01]
FR-F01-02	UC-01	SCR-01	[TBD: PA3 AuthMiddleware / JWT RBAC]	[TBD: PA4 TC-F01-02]
FR-F01-04	UC-02	SCR-02	[TBD: PA3 UserController / REST endpoints]	[TBD: PA4 TC-F01-04]
FR-F02-01	UC-03	SCR-03	[TBD: PA3 BookCatalogController / POST /books]	[TBD: PA4 TC-F02-01]
FR-F02-03	UC-03	SCR-03	[TBD: PA3 BookCatalogController / DELETE /books/:id]	[TBD: PA4 TC-F02-03]
FR-F03-01	UC-04	SCR-04	[TBD: PA3 BookCopyController / POST /copies]	[TBD: PA4 TC-F03-01]
FR-F03-03	UC-04	SCR-04	[TBD: PA3 BookCopy state machine schema]	[TBD: PA4 TC-F03-03]
FR-F04-01	UC-05	SCR-05	[TBD: PA3 SearchController / GET /books/search]	[TBD: PA4 TC-F04-01]
FR-F04-03	UC-06	SCR-06	[TBD: PA3 BookDetailController / GET /books/:id]	[TBD: PA4 TC-F04-03]
FR-F05-01	UC-07	SCR-07	[TBD: PA3 LoanController / POST /loans/checkout]	[TBD: PA4 TC-F05-01]
FR-F05-04	UC-08	SCR-08	[TBD: PA3 LoanController / POST /loans/return]	[TBD: PA4 TC-F05-04]
FR-F05-06	UC-09	SCR-11	[TBD: PA3 LoanController / POST /loans/:id/renew]	[TBD: PA4 TC-F05-06]
FR-F06-01	UC-10	SCR-09	[TBD: PA3 ReservationController / POST /reservations]	[TBD: PA4 TC-F06-01]
FR-F06-04	UC-11	SCR-10	[TBD: PA3 Reservation hold timer & routing]	[TBD: PA4 TC-F06-04]
FR-F07-01	UC-12	SCR-11	[TBD: PA3 DashboardController / GET /reader/loans]	[TBD: PA4 TC-F07-01]

FR-F07-03	UC-12	SCR-12	[TBD: PA3 OverdueController / GET /librarian/overdue]	[TBD: PA4 TC-F07-03]

10 Requirements Review

The formal requirements review was conducted through TA Consultation (SES-01 on 2026-11-01) and Team Walkthrough (SES-02 on 2026-11-02). Identified issues were resolved and approved to establish the frozen requirements baseline:

#	Issue found	Requirement(s)	Resolution
1	Undefined reader loan limit and borrowing period in PA1 proposal baseline	**FR-F05-02, BR-01, BR-02**	Resolved during PA2 Kickoff (Decision D-PA2-01): Capped at 3 concurrent books per reader and 14 calendar days borrowing duration with max 1 renewal.
2	Undefined reservation hold pickup window duration upon copy return	**FR-F06-04, BR-08**	Resolved during PA2 Kickoff (Decision D-PA2-02): Fixed hold pickup window at 72 hours (3 calendar days) before automated queue advancement or release.
3	Potential concurrency race conditions during simultaneous checkouts of last copy or hold routing	**FR-F05-01, FR-F06-04, NFR-05**	Resolved during PA2 Kickoff (Decision D-PA2-03 & Q1 input): Mandated PostgreSQL ACID transactions with SELECT ... FOR UPDATE row-level locks.

11 AI Usage Log

AI use for this consolidated draft is disclosed below. The responsible team member must confirm ownership, check all outputs, edit any inaccuracies, and record any other team members' significant AI use before submission. No human acceptance test or formal approval is implied by the drafting process.

#	Member	Tool	Task / purpose	How the output was verified and changed
1	Nguyễn Phan Thế Vinh (Team Lead / PM · Tester)	Claude 3.5 Sonnet, ChatGPT (GPT-4o)	PA2 Kickoff, open items resolution (loan limit, pickup window), Master ID Registry freeze (WBS 2.1.1 / Task PA2-V1), Acceptance Criteria drafting (AC-F01-01 to AC-F07-02), and RTM schema setup.	Verified against PA0/PA1 scope and team contract; confirmed by all 4 members during kickoff meeting; reconciled 32 FRs, 8 NFRs, 10 BRs, 12 UCs, 12 SCRs, 30 ACs to prevent numbering overlaps and unblock parallel authoring.
2	Tran Ho Anh Quan	ChatGPT	PA2 technical feasibility and implementation risk review (Q1-Q6), recorded in docs/ai-usage/pa2-q1-q6-quan.md.	Source log states review checked against frozen registry, kickoff decisions and repository tree; ChatGPT usage confirmed for this revision
3	Nguyen Thanh Thao Nguyen	ChatGPT (OpenAI, GPT-6)	Assisted with PA2 Business Analysis tasks (N1–N6), including domain question identification, glossary and problem statement drafting, functional requirements (F01–F07), business rules (BR-01 to BR-10), and use case specifications (UC-01, UC-03, UC-05, UC-10, UC-11). Supported requirements organization, flow descriptions, and terminology consistency.	Cross-checked drafts against the PA1 scope, PA2 SRS template, frozen ID Registry, and Figma frames. Revised login, catalog, and reservation flows to improve consistency with the UI designs. Preserved approved FR/BR/UC identifiers and marked unresolved requirements as TBD. Final baseline approval and review issue closure require team confirmation.
4	Vu Thi Anh	Claude Sonnet 5.5, ChatGPT, Figma Make, & Google Gemini Agent	Figma UI wireframe design (SCR-01..12), interactive prototype, UC-12 spec, A11y audit, and UI/UC alignment review (Tasks PA2-A1 to A6)	Utilized Claude Sonnet 5.5, ChatGPT, and Figma Make for initial layout & component suggestions; manually structured and refined all Figma frames to strictly enforce Master ID Registry & SRS business rules; adjusted text contrast (>= 4.5 : 1) and added missing UI error/hold states; configured prototype interactions and verified public access in incognito mode; exported 12 final PNG screenshots.

Project source references (for reviewer cross-check):

UniLib GitHub repository

Frozen SRS ID registry

Functional requirements

Business rules

Acceptance criteria

Figma design